Probability that A wins a point, given that A serves $= p$

Probability that B wins a point, given that B serves $= q$

$0 \rightarrow 15 \rightarrow 30 \rightarrow 40 \rightarrow$ Win
(1) (2) (3) (4)
(Margin at least 2 points)

Deuce at 40 - 40
(3) (3)

→ Inside A Game

1) Probability A wins 4-0

if A serves $= {}^4C_4\, p^4$

if B serves $= {}^4C_4\, (1-q)^4$

2) Probability A wins 4-1

if A serves $= {}^5C_4\, p^4(1-p) - [1 \times p^4 \times (1-p)]$

↙ Selecting 4 point sequences out of 5 in which A wins

↓ Subtracting the case in which A wins first 4 and loses last

$= 4p^4(1-p)$

if B serves $= 4(1-q)^4 q$

3) Probability A wins 4-2

if A serves $= {}^6C_4\, p^4(1-p)^2 - {}^5C_4\, p^4(1-p)^2$

Probability A loses the last and last two points

$= 10\, p^4(1-p)^2$

if B serves $= 10(1-q)^4 q^2$

4) Game reaches deuce (3-3), A wins

- Probability of reaching deuce, given A serves $= {}^6C_3 p^3 (1-p)^3$

$= 20 p^3 (1-p)^3$

- After deuce:

A B A B ....
B A B A ....

$\underbrace{\quad\quad\quad\quad\quad\quad\quad\quad}_{m \text{ point sequences}} \quad \underbrace{A\ A}_{\text{A wins last 2}}$

m ↓ even, $0 \le m < \infty$

$P = p^2$

The m point sequences can be divided into $m/2$ point sequences of two points each.

During those two points, A wins 1 point, B wins 1 point.

Probability of that happening, given A serves $= \underbrace{p(1-p)}_{\text{A wins first}} + \underbrace{(1-p)p}_{\text{B wins first}}$

$= 2p(1-p)$

Now, net probability that A wins the game given he serves $= \left[20p^3(1-p)^3\right] \times p^2 \left[\underbrace{\{2p(1-p)\}^0}_{(n=0)} + \underbrace{\{2p(1-p)\}^1}_{n=2} + \ldots \right]$

↑ Am infinite GP

$$\text{Therefore, } P = 20p^3(1-p)^3 \times p^2 \times \frac{1}{1-2p(1-p)}$$

$$\therefore \text{if A serves} = \frac{20p^5(1-p)^3}{1-2p(1-p)}$$

$$\text{if B serves} = \frac{20(1-q)^5 q^3}{1-2q(1-q)}$$

Therefore,

Probability A wins a game, given that A serves:

$$t = p^4 + 4p^4(1-p) + 10p^4(1-p)^2 + \frac{20p^5(1-p)^3}{1-2p(1-p)}$$

Probability A wins a game, given that B serves:

$$r = (1-q)^4 + 4(1-q)^4 q + 10(1-q)^4 q^2 + \frac{20(1-q)^5 q^3}{1-2(1-q)q}$$

→ Inside a Set (Assuming A serves during First Game)

A can win 6-0, 6-1, 6-2, 6-3, 6-4

7-5 → if score reached 5-5

7-6 → if score reached 6-6 after reaching 5-5

1) A wins 6-0

$$P_{6-0} = t^3 r^3$$ [A serves 3 sets, B serves 3 sets]

2) A wins 6-1

Victory: — — — — — — A

Serve A B A B A B A

Prob t

In the remaining 6 sets, score is 5-1
(3A serves, 3B serves)

Cases:

(i) A wins 3 A serves, 2 B serves

$$P = \left({}^3C_3\, t^3\right)\left[(1-r)\cdot r^2 \times {}^3C_2\right]$$

(ii) A wins 2A serves, 3B serves

$$P = \left[{}^3C_2\, t^2(1-t)\right]\left[{}^3C_3\, r^3\right]$$

∴ Net probab of A winning set 6-1

$$P_{6-1} = 3t^4r^2(1-r) + 3t^3r^3(1-t)$$

3) A wins set 6-2

A wins last game with prob. r

In remaining 7 sets, score is 5-2
(4 A serves, 3 B serves)

Cases:

(i) A wins 4A serves, 1B serve

$$P = \left[{}^4C_4\, t^4\right]\left[{}^3C_1\, r(1-r)^2\right]$$

(ii) A wins 3A serves, 2B serves

$$P = \left[{}^4C_3\, t^3(1-t)\right]\left[{}^3C_2\, r^2(1-r)\right]$$

(iii) A wins 2 A serves, 3B serves

$$P = \left[{}^4C_2\, t^2(1-t)^2\right]\left[{}^3C_3\, r^3\right]$$

∴ Net probab of A winning set 6-2

$$P_{6-2} = 3t^4r^2(1-r)^2 + 12t^3r^3(1-r)(1-t) + 6t^2r^4(1-t)^2$$

4) A wins a set 6-3

Wim:	—	—	—	—	—	—	—	—	A
Serve:	A	B	A	B	A	B	A	B	A
Bob:									t

In remaining 8 sets, score is $5^{A}-3^{B}$

(4A serves, 4B serves)

Cases:

(i) A wins 4A serves, 1B serve

$$P = [{}^4C_4\ t^4][{}^4C_1\ r\ (1-r)^3]$$

(ii) A wins 3A serves, 2B serves

$$P = [{}^4C_3\ t^3(1-t)][{}^4C_2\ r^2\ (1-r)^2]$$

(iii) A wins 2A serves, 3B serves

$$P = [{}^4C_2\ t^2(1-t)^2][{}^4C_3\ r^3\ (1-r)]$$

(iv) A wins 1A serve, 4B serves

$$P = [{}^4C_1\ t\ (1-t)^3][{}^4C_4\ r^4]$$

Net probab of A winning a set 6-3, P_{6-3}

$$= 4t^5 r\ (1-r)^3 + 24 t^4 r^2\ (1-r)^2(1-t) + 24 t^3 r^3\ (1-r)(1-t)^2 + 4 t^2 r^4\ (1-t)^3$$

5) A wins a set 6-4

win:	—	—	—	—	—	—	—	—	—	A
serve:	A	B	A	B	A	B	A	B	A	B
prob:										r

In the remaining 9 sets, score is 5-4

(5 A serves, 4 B serves)

Cases:

(i) A wins 5A serves, 0B serves

$$P = [{}^5C_5\, t^5][{}^4C_0 (1-r)^4]$$

(ii) A wins 4A serves, 1B serve

$$P = [{}^5C_4\, t^4 (1-t)][{}^4C_1\, r\, (1-r)^3]$$

(iii) A wins 3A serves, 2B serves

$$P = [{}^5C_3\, t^3 (1-t)^2][{}^4C_2\, r^2 (1-r)^2]$$

(iv) A wins 2A serves, 3B serves

$$P = [{}^5C_2\, t^2 (1-t)^3][{}^4C_3\, r^3 (1-r)]$$

(v) A wins 1A serve, 4B serves

$$P = [{}^5C_1\, t (1-t)^4][{}^4C_4\, r^4]$$

Net probab of A winning a set 6-4

P_{6-4}

$$= t^5 r (1-r)^4 + 20 t^4 r^2 (1-r)^3 (1-t) + 60 t^3 r^3 (1-r)^2 (1-t)^2 + 40 t^2 r^4 (1-r)(1-t)^3 + 5 t r^5 (1-t)^4$$

6) Special Case: Game score reaching 5-5

10 sets, 5 B serves and 5 A serves

Cases:

(i) A wins 5 A serves, 0 B serves

$$P = [{}^5C_5\, t^5][{}^5C_0\,(1-r)^5]$$

(ii) A wins 4A serves, 1B serve

$$P = [{}^5C_4\, t^4(1-t)][{}^5C_1\, r(1-r)^4]$$

(iii) A wins 3 A serves, 2B serves

$$P = [{}^5C_3\, t^3(1-t)^2][{}^5C_2\, r^2(1-r)^3]$$

(iv) A wins 2A serves, 3B serves

$$P = [{}^5C_2\, t^2(1-t)^3][{}^5C_3\, r^3(1-r)^2]$$

(v) A wins 1A serve, 4 B serves

$$P = [{}^5C_1\, t(1-t)^4][{}^5C_4\, r^4(1-r)]$$

(vi) A wins 0A serves, 5B serves

$$P = [{}^5C_0\,(1-t)^5][{}^5C_5\, r^5]$$

$\therefore$ Net probab of game score reaching 5-5 $(P_{5\text{-}5})$

$$= t^5(1-r)^5 + 25t^4r(1-r)^4(1-t) + 100t^3r^2(1-r)^3(1-t)^2 + 100t^2r^3(1-r)^2(1-t)^3 + 25tr^4(1-r)(1-t)^4 + r^5(1-t)^5$$

7) A wins a set 7-5

Score reaches 5-5, A wins 11th game (A served)
A wins 12th game (B served)

∴ Probab of A winning a set 7-5 = $rt(P_{5-5})$

P_{7-5}

8) A wins a set 7-6

i) Score reaches 6-6:

$$P_{6-6} = P_{5-5}\left[t(1-r) + (1-t)r\right]$$

↓ If A wins his serve, loses B's serve

↓ If A loses his serve, wins B's serve

ii) Decider set:

Since the set started with A's serve, A will serve first, followed by 2 B serves, then 2A serves and so on.

Serve	A	B	B	A	A	B	...

Cases:

a) A wins 7-0:

$$P_a = p^3(1-q)^4$$

b) A wins 7-1:

Win								A
Serve	A	B	B	A	A	B	B	A
Prob								p

$$P_b = p\left[{}^3C_1(1-p)p^2 \times {}^4C_4(1-q)^4 + {}^3C_0 p^3 \times {}^4C_1 q(1-q)^3\right]$$

(c) A wins 7-2

win									A
serve	A	B	B	A	A	B	B	A	A
Prob									p

Score 6-2, 4A serves and 4B serves

Cases:

- A loses 2 of his serves

$$P_1 = \left[{}^4C_2 (1-p)^2 p^2\right]\left[{}^4C_0 (1-q)^4\right]$$

- A loses 1A serve, 1B serve

$$P_2 = \left[{}^4C_1 (1-p) p^3\right]\left[{}^4C_1 q (1-q)^3\right]$$

- A loses 2 B serves

$$P_3 = \left[{}^4C_0 p^4\right]\left[{}^4C_2 q^2 (1-q)^2\right]$$

$$P_c = \text{Prob} = p(P_1 + P_2 + P_3)$$

d) A wins 7-3

win										A
serve	A	B	B	A	A	B	B	A	A	B
prob										$1-q$

Score 6-3, 5A serves, 4B serves

Cases:

- A loses 3 of his serves

$$P_1 = \left[{}^5C_3 (1-p)^3 p^2\right]\left[{}^4C_0 (1-q)^4\right]$$

- A loses 2 of his serves, 1B

$$P_2 = \left[{}^5C_2 (1-p)^2 p^3\right]\left[{}^4C_1 q (1-q)^3\right]$$

- A loses 1A, 2B

$$P_3 = \left[{}^5C_1 (1-p) p^4\right]\left[{}^4C_2 q^2 (1-q)^2\right]$$

- A loses 0A, 3B

$$P_4 = \left[{}^5C_0 p^5\right]\left[{}^4C_3 q^3 (1-q)\right]$$

$$P_d = (1-q)(P_1 + P_2 + P_3 + P_4)$$

e) A wins 7-4

win	—	—	—	—	—	—	—	—	—	—	A
serve	A	B	B	A	A	B	B	A	A	B	B
prob											(1-q)

score 6-4 , 5A serves, 5B serves

Cases:

- A loses 4 A serves, 0B

$$P_1 = \left[{}^5C_4 (1-p)^4 p\right]\left[{}^5C_0 (1-q)^5\right]$$

- A loses 3 A, 1B

$$P_2 = \left[{}^5C_3 (1-p)^3 p^2\right]\left[{}^5C_1 (1-q)^4 q\right]$$

- A loses 2A, 2B

$$P_3 = \left[{}^5C_2 (1-p)^2 p^3\right]\left[{}^5C_2 (1-q)^3 q^2\right]$$

- A loses 1A, 3B

$$P_4 = \left[{}^5C_1 (1-p) p^4\right]\left[{}^5C_3 (1-q)^2 q^3\right]$$

- A loses 0A, 4B

$$P_5 = \left[{}^5C_0 p^5\right]\left[{}^5C_4 (1-q) q^4\right]$$

$$P_e = \text{prob} = (1-q)(P_1 + P_2 + P_3 + P_4 + P_5)$$

f) A wins 7-5

win	—	—	—	—	—	—	—	—	—	—	—	A
serve	A	B	B	A	A	B	B	A	A	B	B	A
prob												p

score 6-5 , 5 A serves, 6 B serves

Cases:

- A loses 5A serves, 0B

$$P_1 = \left[{}^5C_5 (1-p)^5\right]\left[{}^6C_0 (1-q)^6\right]$$

- A loses 4A, 1B

$$P_2 = \left[{}^5C_4 (1-p)^4 p\right]\left[{}^6C_1 (1-q)^5 q\right]$$

- A loses 3A, 2B

$$P_3 = \left[{}^5C_3 (1-p)^3 p^2\right]\left[{}^6C_2 q^2 (1-q)^4\right]$$

- A loses 2A, 3B

$$P_4 = \left[{}^5C_2 (1-p)^2 p^3\right]\left[{}^6C_3 q^3 (1-q)^3\right]$$

- A loses 1A, 4B

$$P_5 = \left[{}^5C_1 (1-p) p^4\right]\left[{}^6C_4 q^4 (1-q)^2\right]$$

- A loses 0A, 5B

$$P_6 = \left[{}^5C_0 p^5\right]\left[{}^6C_5 q^5 (1-q)\right]$$

$$P_6 = \text{prob} = p\left[P_1 + P_2 + P_3 + P_4 + P_5 + P_6\right]$$

g) Deuce

Score first reaches 6-6, then A needs a two point lead to win.

* For reaching 6-6

A	B	B	A	A	B	B	A	A	B	B	A

6A serves, 6B serves

- A wins 6A, 0B

$$P_1 = \left[{}^6C_6 p^6\right]\left[{}^6C_0 q^6\right]$$

- A wins 5A, 1B

$$P_2 = \left[{}^6C_5 p^5 (1-p)\right]\left[{}^6C_1 (1-q) q^5\right]$$

- A wins 4A, 2B

$$P_3 = \left[{}^6C_4 p^4 (1-p)^2\right]\left[{}^6C_2 (1-q)^2 q^4\right]$$

- A wins 3A, 3B

$$P_4 = \left[{}^6C_3 p^3 (1-p)^3\right]\left[{}^6C_3 (1-q)^3 q^3\right]$$

- A wins 2A, 4B

$$P_5 = \left[{}^6C_2\, p^2 (1-p)^4\right]\left[{}^6C_4\, q^2 (1-q)^4\right]$$

- A wins 1A, 5B

$$P_6 = \left[{}^6C_1\, p (1-p)^5\right]\left[{}^6C_5\, q (1-q)^5\right]$$

- A wins 0A, 6B

$$P_7 = \left[{}^6C_0 (1-p)^6\right]\left[{}^6C_6 (1-q)^6\right]$$

$$P_{6\text{-}6} = P_1 + P_2 + P_3 + P_4 + P_5 + P_6 + P_7$$

Post 6-6

win	A B	A B				A A
Serve	A A	B B	A A	B B	...	A A / B B

m pairs

if m = odd → ends with BA ↳ Prob $= (1-q)p$

$$P_o' = \left[pq + (1-p)(1-q)\right]^1 + \left[\ldots\right]^3 + \ldots$$

$$\therefore P_o' = \frac{pq + (1-p)(1-q)}{1 - \left[pq + (1-p)(1-q)\right]^2}$$

if m = odd → serve ends with AB ↳ Prob $= p(1-q)$

$$P_e' = 1 + \left[pq + (1-p)(1-q)\right]^2 + \left[\ldots\right]^4$$

Same expressions

↓ Can be merged

$$\therefore P_{\text{post } 6\text{-}6} = \frac{p(1-q)}{1 - \left[pq + (1-p)(1-q)\right]}$$

∴ Probability of A winning after tie in tie-break game, P_g $= P_{66} \times \dfrac{p(1-q)}{1 - \left[pq + (1-p)(1-q)\right]}$

Probability of winning a set 7-6 (P_{7-6}) $= P_{66}\left[P_a + P_b + P_c + P_d + P_e + P_f + P_g + P_h\right]$

Probability of A winning a set, given set started with his serve $= P_{6-0} + P_{6-1} + P_{6-2} + P_{6-3} + P_{6-4} + P_{7-5} + P_{7-6}$